Digital Signal Processing Course Project

Every team must implement **BOTH** projects. Each project contributes **10%** to the course grade.

I– ECG Signal Denoising for Telemedicine Applications (10 Points)

Objective

Design and implement digital filters to remove noise from Electrocardiogram (ECG) signals used in remote health monitoring and wearable medical systems.

Signal Characteristics

Typical ECG properties:

- Sampling frequency: 360 Hz
- Useful ECG bandwidth: 0.5 – 100 Hz

Common noise components:

- Baseline wander (< 0.5 Hz) due to respiration
- Power-line interference at 50 Hz
- Muscle (EMG) noise between 20 – 150 Hz

ECG Data Source

Students must use ECG recordings from the publicly available **MIT-BIH Arrhythmia Database** hosted on PhysioNet.

The database contains 48 annotated ECG recordings sampled at 360 Hz and widely used in biomedical signal processing research.

Students may download the database from:

- `https://physionet.org/content/mitdb/1.0.0/`

For consistency across teams, students are encouraged to use the following records:

- **Record 100** (primarily **normal** sinus rhythm)

- **Record 106** (contains **abnormal** ventricular ectopic beats)

Each record includes three files:

- `.dat` – ECG signal samples
- `.hea` – signal header information
- `.atr` – beat annotations

Students can extract a segment of the signal for filter design, analysis, and visualization.

Problem Definition and Specifications (2 Points)

Students must:

1. Describe ECG signal characteristics & clearly define the filtering objectives (0.5)
2. Identify the required filter types: (0.5)
   - High-pass filter (baseline removal)
   - Notch filter at 50 Hz
   - Low-pass filter (muscle noise reduction)
3. Define and justify filter specifications: (0.5)
   - Passband ripple
   - Stopband attenuation
   - Transition bandwidth
   - Cutoff frequencies
4. Identify realistic implementation constraints such as: (0.5)
   - Maximum filter order
   - Computational complexity
   - Latency
   - Phase distortion limitations

Filter Design and Analysis (4 Points)

Students must design at least three different digital filters, for example:

- FIR (Window-based method)
- IIR Butterworth
- IIR Chebyshev (Type I or II)

For each design students must:

- Specify filter type and design method (0.5)
- Provide filter coefficients (0.5)
- Plot magnitude and phase responses (1)
- Plot impulse and step responses (1)
- Show pole-zero diagram (0.5)
- Verify compliance with specifications (0.5)

Implementation and Validation (4 Points)

- Apply the designed filters to ECG data. (1)
- Compare the ECG signal before and after filtering. (1)
- Demonstrate effectiveness using: (1.5)
  - Power spectral density (Welch method)
  - Spectrogram (STFT)
  - Quantitative indicators such as SNR improvement
- Provide time-domain waveform comparison. (0.25)
- Discuss distortion and possible limitations. (0.25)

II– Multi-Band Speech Equalizer for Podcast Enhancement (10 Points)

Objective

Develop a MATLAB-based multi-band digital equalizer for speech clarity enhancement in podcasts, recorded lectures, and voice recordings.

Program Requirements

The program must allow the user to input:

- Audio file name
- Gain of each frequency band (in dB)
- Filter type (FIR or IIR)
- Filter order
- Output sample rate

Modes of Operation

1 – Preset Mode (Speech-Optimized Bands)

Predefined frequency bands:

- 0–100 Hz
- 100–300 Hz
- 300–800 Hz
- 800–2 kHz
- 2–5 kHz
- 5–10 kHz
- 10–20 kHz

2 – Custom Mode

- Minimum 5 bands
- Maximum 10 bands
- First band must start at 0 Hz
- Last band must end at 20 kHz

Filter Design Options

Students may design the equalizer filters using either FIR or IIR structures.

FIR Filters

- Window types:
  - Hamming
  - Hanning
  - Blackman
- User-defined or default filter order

IIR Filters

- Butterworth
- Chebyshev Type I
- Chebyshev Type II
- User-defined or default filter order

Filter Design and Analysis (3 Points)

Students must generate for **each band**:

- Magnitude response (0.5)
- Phase response (0.5)
- Impulse response (0.5)
- Step response (0.5)
- Filter order (0.5)
- Pole-zero plot (0.5)

Signal Processing Steps (5 Points)

1. Design band filters. (1)
2. Filter the input signal. (0.5)
3. Apply gain (in dB). (0.5)
4. Combine filtered bands in the time domain. (0.75)
5. Compare original and processed signals in time & frequency domians (0.75)
6. Plot Power Spectral Density. (0.5)
7. Generate spectrogram. (0.5)
8. Play and save the output file. (0.25)
9. Demonstrate: (0.25)
    - FIR implementation
    - IIR implementation
    - Output sample rate multiplied by four
    - Output sample rate reduced to half

Performance Evaluation (2 Points)

Students must:

- Compare spectra before and after equalization
- Analyze spectrogram differences
- Perform listening evaluation
- Discuss artifacts, distortion, and trade-offs

Team Formation and Submission Requirements

- Each team must consist of 2 or 3 students.
- Both projects must be implemented.
- Deadline to form teams: 15 March.
- Project submission deadline: 15 May.
- Submission must include:
  - One PDF report
  - All **MATLAB** codes (well-commented)
  - Any used data files
- The report must include:
  - Problem formulation
  - Filter specifications
  - Design procedures
  - All plots
  - Sample runs
  - Results for modified sample rates
  - Discussion and insights
- A project discussion session will be conducted. Students must demonstrate full understanding of their implementation.

Important Notes

- **Discussion:** Students must attend the project discussion. Evaluation will depend on task quality, documentation, and the ability to explain the implementation.

Evaluation Level	Score
Task is not done	1
Inadequate	2
Needs improvement	3
Outstanding	4

  Outstanding means the task meets expectations and the discussion answers demonstrate clear understanding.
- **Documentation:** Code must be well commented and variables meaningfully named.
- **Deliverables:** All materials must be submitted via Classroom before the deadline. Only one member should submit the work with the group number.